

INFORMATICS INSTITUTE OF TECHNOLOGY

Foundation certificate in higher education.

Module: DOC334 – Computer Programming

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i. Table of Contents

ii. List of figure ii
2. Solution 3
2.1 dm.py 3
2.2 Sub.py 6
3. Test cases 11

ii. List of figure

Figure 1 Trishaw Price Plan under DropMe™ 1
Figure 2 Command types 1
Figure 3 Sample test cases 2
Figure 4 dm.py Alvin Zuhar /pro12 11
Figure 5 dm.py Alvin Razi 11
Figure 6 dm.py eji oofr ff ff alvin 11
Figure 7 dm.py maali jjmz 12
Figure 8 dm.py mali JaMz /pro17 12
Figure 9 dm.py mali zuhar /pro4 /v 12
Figure 10 dm.py razi jamz /c 13
Figure 11 dm.py zuhar zuhar /c 13
Figure 12 dm.py 13

1. Description of the problem statement

City Name	*Alvin*	*Jamz*	*Razi*	*Mali*	*Zuhar*
Alvin	0	20	40	40	20
Jamz	20	0	20	40	40
Razi	40	20	0	20	40
Mali	40	40	20	0	20
Zuhar	20	40	40	20	0

Figure 1 Trishaw Price Plan under DropMe™

The "The Kingdom of Miranda" is a circular shaped small island with 5 main cities named as Alvin , Jamz, Razi, Mali and Zuhar.The official currency of the country is KMD (Kingdom of Miranda Dollars). The country uses a single cab service known as DropMe™. The DropMe™ price chart for travelling is shown below. All the prices are in KMD.

Based on the above chart, the price to travel from Alvin to Jamz is just 20 KMD while Razi to Razi is 0 KMD; which means the ride is free within the same city. The country uses only Trishaws, Cars, and Vans for cab services. The default transport is Trishaws while passengers can ask for cars or vans if they prefer. Car prices have doubled while van prices have tripled. For example, a Trishaw ride from Jamz to Mali is 40 KMD while the same ride in a car is 80 KMD. It will be 120 KMD if you take a van. The vehicles will always take the shortest path for travelling. The president of The Kingdom of Miranda hired a good Python coder from IIT (You!) to develop a console application which will allow users to view trip details and generate invoices. The application is used by the passenger. The following commands can be issued to run the application from the client side.

Command Types

Command	Description
C:\Users\Nishan>dm alvin razi	Shows the price between the two cities and generates an invoice file for the trip
C:\Users\Nishan>dm alvin razi /pro2	Shows the price between the two cities after applying the promo code. The promo code amount is deducted from the final bill value. This will generate an invoice file for the trip with promo details. 2 KMD is reduced here.
C:\Users\Nishan>dm /price	Show the full price plan for the whole country (for all 3 vhicles)
C:\Users\Nishan>dm alvin razi /c	Shows the price between the two cities and generates an invoice file for the trip. This uses a car (/c for car while /v for a van. Default is Trishaw)
C:\Users\Nishan>dm alvin razi /pro10 /v	Shows the price between the two cities while apply a 10 KMD reduction to total bill. The rider prefers a van.
C:\Users\Nishan>dm /?	Show how this "dm" command functions

Figure 2 Command types

The promo codes are available under especial circumstances or issued individually. The format of these codes is as below.

- pro2 (2 KMDs are reduced from the total bill)
- pro5 (5 KMDs are reduced from the total bill)
- pro10 (10 KMDs are reduced from the total bill)

The above three are samples while the promotion amounts can range from 1 KMD to maximum of 15 KMD.

The application itself will generate price reductions for customers who use the application. These auto-generated promotions are always at a fixed price of 5 KMD. These are not taking place if the passenger uses a promotional code. This feature is a random feature! Only the lucky passengers will get this special promotion time-to-time. Every time a passenger uses the app, the invoice shown will be saved in a text file in the format. Three such sample files are shown below.

Date : 2023-06-26 Time: 06:28:29 Start : Alvin End : Razi Amount : 40 KMD Promo : 0 Random Reduction : 0 Final payment : 40 KMD	Date : 2023-06-26 Time: 06:30:29 Start : Alvin End : Razi Amount : 40 KMD Promo : 10 KMD Random Reduction : 0 Final payment : 30 KMD	Date : 2023-06-27 Time: 05:28:00 Start : Alvin End : Razi Amount : 40 KMD Promo : 0 Random Reduction : 5 KMD Final payment : 35 KMD

Figure 3 Sample test cases

A file is saved for each invoice and the file name should look like 2023-06-26 06_30_29_4526.txt

2. Solution

2.1 dm.py

```
# Importing the Calculation holders
import sys
import Calculations.Sub

# Defining the price table
price_chart = {
   'Alvin': {'Alvin': 0, 'Jamz': 20, 'Razi': 40, 'Mali': 40, 'Zuhar': 20},
   'Jamz': {'Alvin': 20, 'Jamz': 0, 'Razi': 20, 'Mali': 40, 'Zuhar': 40},
   'Razi': {'Alvin': 40, 'Jamz': 20, 'Razi': 0, 'Mali': 20, 'Zuhar': 40},
   'Mali': {'Alvin': 40, 'Jamz': 40, 'Razi': 20, 'Mali': 0, 'Zuhar': 20},
   'Zuhar': {'Alvin': 20, 'Jamz': 40, 'Razi': 40, 'Mali': 20, 'Zuhar': 0}
}

# Defining the promo codes
promo_codes = {
   'pro2': 2,
   'pro3': 3,
   'pro4': 4,
   'pro5': 5,
   'pro6': 6,
   'pro7': 7,
   'pro8': 8,
   'pro9': 9,
   'pro10': 10,
   'pro11': 11,
   'pro12': 12,
   'pro13': 13,
   'pro14': 14,
   'pro15': 15
}

# Main function to handle the command line inputs
def main() :
   if len(sys.argv) < 2 or len(sys.argv) > 6 :
      print ('\nInvalid Command. There are too many or too few arguments entered by you.')
      return

   command = sys.argv[1:]

   if len(command) == 1 :
      if command[0] == '/?' :
         Calculations.Sub.show_help()

      elif command[0] == '/price' :
         Calculations.Sub.show_price_plan()
```

```
    else:
        print ('\nInvalid Command. A valid command should be provided.')

else:
    command_parts = command
    num_parts = len(command_parts)

    start_city = command_parts[0].capitalize()
    end_city = command_parts[1].capitalize()

    vehicle = 'Trishaw'
    promo = 0

    if (start_city not in price_chart) or (end_city not in price_chart) :
        print ('\nInvalid Command. A valid city name should be provided.')
        return


    if num_parts == 3 :
        part = command_parts[2]

        if part.startswith('/'):
            if part[1:] in promo_codes:
                promo = promo_codes[part[1:]]

            elif part[1:] == 'c':
                vehicle = 'Car'

            elif part[1:] == 'v':
                vehicle = 'Van'

            else:
                print ('\nInvalid Command. Please Enter a valid vehicle type or valid promo code.')
                return
        else:
            print ('\nInvalid Command. A valid command should be provided.')
            return

    elif num_parts == 4 :
        part_1 = command_parts[2]

        if part_1.startswith('/'):
            if part_1[1:] in promo_codes:
                promo = promo_codes[part_1[1:]]

            else:
                print ('\nInvalid Command. Please Enter A valid promo code.')
                return
        else:
            print ('\nInvalid Command. Please Enter A valid promo code.')
            return
```

```
        part_2 = command_parts[3]

        if part_2.startswith('/'):
            if part_2[1:] == 'c':
                vehicle = 'Car'

            elif part_2[1:] == 'v':
                vehicle = 'Van'

            else:
                print ('\nInvalid Command. Please Enter a valid vehicle type.')
                return
        else:
            print ('\nInvalid Command. Please Enter a valid vehicle type.')
            return

    amount = Calculations.Sub.calculate_price(start_city, end_city, vehicle)
    random_reduction = 0

    if promo == 0:
        random_reduction = Calculations.Sub.generate_random_reduction()
        promo_ending = ''
    else:
        promo_ending = 'KMD'

    final_payment = amount - promo - random_reduction
    Calculations.Sub.generate_invoice(start_city, end_city, amount, promo, random_reduction, vehicle)

    if random_reduction != 0 :
        random_reduction_ending = 'KMD'
    else:
        random_reduction_ending = ''

    print ('\n═══════════ Trip Details ═══════════\n')
    print ('          Start : ' + start_city)
    print ('            End : ' + end_city)
    print ('        Vehicle : ' + vehicle + '\n')
    print ('         Amount : ' + str(amount))
    print ('          Promo : ' + str(promo) + promo_ending)
    print ('Random Reduction : ' + str(random_reduction) + random_reduction_ending)
    print ('   Final Payment : ' + str(final_payment))
    print ('\n═══════════════════════════════════════\n')
    print ('Invoice Created Successfully.')

# Running the main function
main()
```

2.2 Sub.py

```
# Imports Libraries
import random
import datetime

# Define the price chart
price_chart = {
    'Alvin': {'Alvin': 0, 'Jamz': 20, 'Razi': 40, 'Mali': 40, 'Zuhar': 20},
    'Jamz': {'Alvin': 20, 'Jamz': 0, 'Razi': 20, 'Mali': 40, 'Zuhar': 40},
    'Razi': {'Alvin': 40, 'Jamz': 20, 'Razi': 0, 'Mali': 20, 'Zuhar': 40},
    'Mali': {'Alvin': 40, 'Jamz': 40, 'Razi': 20, 'Mali': 0, 'Zuhar': 20},
    'Zuhar': {'Alvin': 20, 'Jamz': 40, 'Razi': 40, 'Mali': 20, 'Zuhar': 0}
}

# Define the vehicle prices
vehicle_prices = {
    'Trishaw': 1,
    'Car': 2,
    'Van': 3
}

# Generate a random reduction for lucky passengers
def generate_random_reduction():
    return random.choice([0, 5])

# Calculate the price for a given trip and vehicle
def calculate_price(start, end, vehicle):
    distance = price_chart[start][end]
    price_per_km = vehicle_prices[vehicle]
```

```
    cost = distance * price_per_km
    return cost

# Generate the invoice file
def generate_invoice(start, end, amount, promo, random_reduction, vehicle):
    current_datetime = datetime.datetime.now()
    date_str = current_datetime.strftime('%Y-%m-%d')
    time_str = current_datetime.strftime('%H_%M_%S_%f')[:-2]
    file_time_str = current_datetime.strftime('%H:%M:%S')
    filename = date_str + ' ' + time_str + '.txt'

    final_payment = amount - promo - random_reduction

    if promo != 0 :
        promo_ending = 'KMD'
    else:
        promo_ending = ''

    if random_reduction != 0 :
        random_reduction_ending = 'KMD'
    else:
        random_reduction_ending = ''

    file = open(filename, 'w')

    file.write('Date : ' + date_str + '\n')
    file.write('Time : ' + file_time_str + '\n')
    file.write('Start : ' + start + '\n')
    file.write('End : ' + end + '\n')
    file.write('Amount : ' + str(amount) + ' KMD' + '\n')
    file.write('Promo : ' + str(promo) + promo_ending + '\n')
```

```
    file.write('Random Reduction : ' + str(random_reduction) + random_reduction_ending + '\n')
    file.write('Final Payment : ' + str(final_payment) + ' KMD')

    file.close()


# Show the full price plan for all cities and vehicles
def show_price_plan():
    print ('\n Price Plan :\n')
    print
('╔═══════════════════════════════════════════════════════════════════════════
═══════════════════════════╗')

    for city, city_prices in price_chart.items():
        print ('║ ' + city.center(73) + ' ║')
        print
('╠═══════════════════════════════════════════════════════════════════════════
═══════════════════════════╣')

        for dest_city, price in city_prices.items():
            trishaw_price = str(price) + ' KMD'
            car_price = str(price * vehicle_prices['Car']) + ' KMD'
            van_price = str(price * vehicle_prices['Van']) + ' KMD'
            print ('║ ' + dest_city.ljust(10) + ' - Trishaw: ' + trishaw_price.ljust(13) + ' Car: ' + car_price.ljust(13)
+ ' Van: ' + van_price.ljust(13) + ' ║')

        print
('╠═══════════════════════════════════════════════════════════════════════════
═══════════════════════════╣')

    print
('╚═══════════════════════════════════════════════════════════════════════════
═══════════════════════════╝')


# Show the help message
```

```
def show_help():
    print ('\n Command Types :\n')
    print
('┌──────────────
```

```
    print ('│                                        │ Shows the price between the two cities          │')
    print ('│ dm.py <start_city> <end_city> /<promo_code> /<vehicle_code>  │ while applying a promo code and using a  │')
    print ('│                                        │ specific vehicle.                               │')
    print ('├──────────────────────────────────────────────────────────────────────────────────────────────┤')
    print ('│ dm.py /?                               │ Shows the instructions for using the "dm"       │')
    print ('│                                        │ command.                                        │')
    print ('└──────────────────────────────────────────────────────────────────────────────────────────────┘')
```

3. Test cases

1. dm.py Alvin Zuhar /pro12.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py Alvin Zuhar /pro12

========= Trip Details =========

          Start : Alvin
            End : Zuhar
        Vehicle : Trishaw

         Amount : 20
          Promo : 12KMD
Random Reduction : 0
   Final Payment : 8

================================

Invoice Created Successfully.
```

Figure 4 dm.py Alvin Zuhar /pro12

2. dm.py Alvin Razi.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py Alvin Razi

========= Trip Details =========

          Start : Alvin
            End : Razi
        Vehicle : Trishaw

         Amount : 40
          Promo : 0
Random Reduction : 0
   Final Payment : 40

================================

Invoice Created Successfully.
```

Figure 5 dm.py Alvin Razi

3. dm.py eji oofr ff ff alvin.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py eji oofr ff ff alvin

Invalid Command. A valid city name should be provided.
```

Figure 6 dm.py eji oofr ff ff alvin

4. dm.py maali jjmz.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py maali jjmz

Invalid Command. A valid city name should be provided.
```

Figure 7 dm.py maali jjmz

5. dm.py mali JaMz /pro17.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py mali JaMz /pro17

Invalid Command. Please Enter a valid vehicle type or valid promo code.
```

Figure 8 dm.py mali JaMz /pro17

6. dm.py mali zuhar /pro4 /v.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py mali zuhar /pro4 /v

========= Trip Details =========

          Start : Mali
            End : Zuhar
        Vehicle : Van

         Amount : 60
          Promo : 4KMD
Random Reduction : 0
   Final Payment : 56

================================


Invoice Created Successfully.
```

Figure 9 dm.py mali zuhar /pro4 /v

7. dm.py razi jamz /c.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py razi jamz /c

========= Trip Details =========

          Start : Razi
            End : Jamz
        Vehicle : Car

         Amount : 40
          Promo : 0
Random Reduction : 5KMD
   Final Payment : 35

================================

Invoice Created Successfully.
```

Figure 10 dm.py razi jamz /c

8. dm.py zuhar zuhar /c.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py zuhar zuhar /c

========= Trip Details =========

          Start : Zuhar
            End : Zuhar
        Vehicle : Car

         Amount : 0
          Promo : 0
Random Reduction : 0
   Final Payment : 0

================================

Invoice Created Successfully.
```

Figure 11 dm.py zuhar zuhar /c

9. dm.py.

```
F:\IIT\Doc 334 Computer Programmong\20222346\Bihandu>dm.py

Invalid Command. There are too many or too few arguments entered by you.
```

Figure 12 dm.py